The Impact of Local Climate Governance on Residential Energy Performance in the UK

An Analysis of Climate Emergency Declarations and Local Area Energy Plans

7th December 2025

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Abstract

This dissertation investigates whether local climate governance initiatives—specifically Climate Emergency Declarations (CEDs) and Local Area Energy Plans (LAEPs)—are associated with measurable improvements in residential energy performance across the United Kingdom. With the residential sector contributing approximately 20% of national CO_2 emissions, understanding the effectiveness of local-level climate action is crucial for achieving Net Zero 2050 targets.

[Add 200-300 words summarizing your research question, methodology, key findings, and conclusions]

Acknowledgements

I would like to express my sincere gratitude to my supervisor, [Supervisor Name], for their invaluable guidance and support throughout this research project. I also thank [other people/institutions] for their assistance with [specific contributions].

[Add personal acknowledgements]

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List of Abbreviations

Abbreviation	**Definition**
CED	Climate Emergency Declaration
CO_2	Carbon Dioxide
EPC	Energy Performance Certificate
GHG	Greenhouse Gas
LAEP	Local Area Energy Plan
MEES	Minimum Energy Efficiency Standards
SAP	Standard Assessment Procedure
UK	United Kingdom

Chapter 1

Introduction

1.1 Background

The residential housing sector is a major contributor to the United Kingdom's carbon footprint, accounting for roughly 20% of national CO_2 emissions (Energy Security and Zero 2023). For this reason, improving the energy performance of homes has become central to the UK's Net Zero 2050 strategy and, more recently, to the more ambitious 2030 targets set by several local authorities. How much energy a home uses depends on its underlying characteristics—its age, type, insulation levels, glazing, and heating system—but also on the policy environment that shapes retrofit activity and minimum performance standards.

Over the past decade, the UK has introduced a series of national policy measures aimed at improving domestic energy performance. These include the Minimum Energy Efficiency Standards (MEES), low-carbon heat incentives, and successive revisions to building regulations. At the same time, a parallel shift has taken place in local governance: since 2019, more than 300 councils have declared a Climate Emergency. Such declarations signal political commitment, but the extent to which they translate into measurable changes in the housing stock remains unclear.

A more structured form of local climate action is emerging through Local Area Energy Plans (LAEPs). These are intended to provide detailed, data-driven pathways for decarbonising heat, buildings, power, and infrastructure. However, only a small number of completed LAEPs currently exist, and there is very limited evidence on their implementation or impact. As a result, it is not yet known whether these local governance tools are associated with observable improvements in residential energy metrics such as EPC ratings or modelled CO_2 emissions.

This uncertainty motivates the present study. Improvements in EPC outcomes over the last decade may reflect national policy cycles, long-term renewal of the housing stock, or, conceivably, early forms of local climate governance. Disentangling these influences is far from straightforward, but a careful empirical

analysis can shed light on the patterns that underlie them.

1.2 Research Questions and Objectives

Focusing on domestic properties in the South West of England, this dissertation addresses the following overarching question:

> *How have national housing energy-policy cycles and emerging local climate governance initiatives related to changes in residential energy performance in the South West, and what role do structural dwelling characteristics play in shaping these outcomes?*

To explore this question, the research pursues five objectives:

1. **Policy trends:** To describe and quantify how EPC efficiency scores and EPC-estimated CO_2 emissions evolved across major national policy phases between 2008 and 2025.
2. **Structural drivers:** To examine how housing characteristics—construction age, main fuel type, property type, and tenure—contribute to variation in energy performance within and across local authorities.
3. **Heterogeneity in policy-period associations:** To assess whether improvements associated with national policy cycles differ across key subgroups, notably between gas and non-gas dwellings and between owner-occupied and private-rented homes.
4. **Local climate governance (descriptive):** To situate these trends relative to the timing of Climate Emergency Declarations and the limited number of LAEPs, acknowledging that available data do not permit causal evaluation of these initiatives.
5. **Predictive modelling:** To develop and assess interpretable machine-learning models that predict EPC outcomes (EPC scores and below/above EPC C status) from a concise set of structural features, thereby indicating how much of the variation in EPC performance can be explained from routinely recorded dwelling attributes.

These objectives span descriptive analysis, fixed-effects regression, heterogeneity investigation, and predictive modelling. Local climate governance is treated with caution and included for contextual interpretation rather than formal causal identification.

1.3 Significance of the Research

The dissertation contributes to the evidence base in three ways.

First, it offers a detailed empirical analysis of long-run residential energy-performance trends using a large dataset of approximately 1.8 million EPC records. By aligning these data with major national policy cycles and estimating fixed-effects regressions for EPC efficiency, modelled CO_2 emissions, and the likelihood of falling below EPC C, the study identifies which policy periods appear most closely associated with improvements.

Second, the modelling framework incorporates rich dwelling-level characteristics and examines how these features interact with policy periods. This helps distinguish between improvements driven by new-build activity and those reflecting deeper changes within the existing stock. The heterogeneity analysis also highlights groups of dwellings—particularly older and non-gas homes—that remain consistently less efficient.

Third, the dissertation develops machine-learning models that predict EPC outcomes from a small set of interpretable features. These models are not intended for operational deployment; rather, they serve to verify the structured nature of EPC scoring and to reinforce the importance of age, tenure, and fuel type as the dominant drivers of energy performance. The findings have practical relevance for targeting retrofit support and informing local energy planning.

Although CEDs and LAEPs cannot be evaluated causally with the available data, the dissertation situates national policy effects within the broader context of growing local climate governance, highlighting where improved data could enhance future evaluation.

1.4 Scope and Limitations

The analysis focuses on domestic EPC lodgements from 2008 to 2025 in the South West of England. This region provides a diverse mix of housing ages, tenures, and heating systems, but the findings may not generalise directly to areas with substantially different stock profiles.

Several limitations arise from the nature of EPC data:

- **Modelled rather than metered performance.** EPC-derived CO_2 emissions and energy-use estimates are based on standardised assumptions and may diverge from real consumption.

- **Event-driven sampling.** EPCs are typically triggered by a sale, rental, or construction event, leading to non-random sampling over time.
- **Local policy data constraints.** Information on CEDs and LAEPs is limited to announcement years or plan publication dates, with no systematic data on implementation, funding, or delivery. For this reason, local climate policies are interpreted descriptively rather than causally.
- **Unobserved physical features.** Important characteristics such as insulation thickness, thermal bridging, or boiler efficiency are not available, constraining both regression and machine-learning models.

Given these constraints, the dissertation does not claim precise causal estimates of specific policies. Instead, it seeks to identify robust associations across national policy cycles and to quantify the structural factors that most strongly shape residential EPC performance.

Chapter 2

Literature Review

2.1 Introduction

This chapter reviews the evidence base relevant to understanding how national housing policies, local climate governance, and structural dwelling characteristics shape residential energy performance. Five strands of literature are examined: (1) the rise of Climate Emergency Declarations (CEDs), (2) the development of Local Area Energy Plans (LAEPs), (3) local-authority capacity and governance challenges, (4) determinants of household energy use and EPC performance, and (5) the complexity of the UK's housing-regulation landscape. Together, these strands highlight substantial gaps in the empirical evaluation of local climate initiatives and provide the conceptual motivation for the analytical framework adopted in this dissertation.

2.2 Climate Emergency Declarations in the UK

Since 2018, more than 75% of UK local authorities have declared a Climate Emergency, signalling political commitment to accelerated climate action. A systematic review by Gudde et al. (2021) analysed public documents from over 300 UK authorities and found substantial variability in the scope, timelines, and governance arrangements behind these declarations. Although political consensus on the urgency of climate action was widespread, most authorities lacked detailed delivery plans, clear funding pathways, or defined roles for citizen engagement.

Crucially, Gudde et al. (2021) identify an "implementation gap": declarations are symbolically powerful but rarely accompanied by operational strategies capable of driving measurable improvements in building emissions. This directly connects to the present research: if CEDs do not consistently translate into structured interventions—such as retrofit programmes, enforcement of minimum standards, or heat decarbonisation schemes—their impact on EPC trends is likely to be limited or highly uneven. This is particularly important because the EPC dataset

used in this dissertation contains no information on implementation intensity, funding, or retrofit delivery at the local level.

The Manchester Climate Emergency Report (Manchester City Council 2019) exemplifies this broader pattern. It sets ambitious targets (net zero by 2038) and highlights areas for action, but provides descriptive rather than empirical evidence of changes in residential energy efficiency. This reinforces the need for caution in attributing changes in EPC outcomes to local declarations alone.

2.3 Local Area Energy Plans: Purpose, Potential, and Constraints

Local Area Energy Plans (LAEPs) have emerged as one of the most promising tools for local decarbonisation planning. Defined by the Energy Systems Catapult as "a data-driven, whole-system approach" that identifies cost-effective pathways to local net zero (Energy Systems Catapult 2021), LAEPs aim to provide detailed modelling of heat decarbonisation options, infrastructure needs, and retrofit trajectories.

However, the ESC's 2021 review reveals that LAEP development in the UK remains at an early stage: at the time of publication, only three authorities had completed a full LAEP (Energy Systems Catapult 2021). Moreover, the report highlights that many councils lack the staffing, engineering expertise, and budgetary capacity to execute such plans effectively:

> "Local authorities do not typically have resources—people or money—to undertake producing LAEPs and delivering the actions that they identify." (Energy Systems Catapult 2021, p. 7)

This observation is echoed by Poulter et al. (2025), who describe a "governance gap" across local and regional energy planning in Great Britain. They argue that roles and responsibilities are fragmented across multiple actors—local councils, distribution network operators, industrial stakeholders, and national regulators—resulting in inconsistent planning quality and limited progress toward implementation. The lack of alignment across these bodies limits the ability of local areas to accelerate toward net zero and reduces the likelihood of observable short-term impacts on residential energy performance indicators such as EPC ratings.

Given this context, the present dissertation treats LAEPs descriptively. The available data record the existence and year of publication of a LAEP but provide no metrics of implementation, funding, or retrofit delivery. As such, LAEPs

cannot be subjected to causal identification within a fixed-effects framework.

2.4 Determinants of Domestic Energy Performance

While local climate governance remains under-evaluated, the determinants of energy use within households are well established. The 2024 *Energies* study (Sewell et al. 2024) provides one of the most comprehensive UK-specific analyses of household electricity and gas consumption. Several findings from this paper directly inform the modelling decisions in this dissertation.

2.4.1 Structural dwelling characteristics

The authors identify *property type* (e.g., detached vs. flat) as the strongest predictor of both gas and electricity use across all household groups. Other high-importance features include:

- **Construction age** — which reflects historical building standards and insulation levels.
- **Heating system and main fuel type** — gas vs. electric systems drive large differences in energy consumption.
- **Tenure** — homeowners exhibit lower gas demand than private renters, partly due to greater autonomy to invest in retrofit.
- **Wall construction** and building fabric measures.

These structural drivers align closely with the explanatory variables included in this dissertation's fixed-effects regressions and predictive modelling.

2.4.2 Limitations of EPC ratings

A key finding from Sewell et al. (2024) is that EPC ratings themselves are not strong predictors of real energy consumption:

> "Current Energy Efficiency (EPC rating) is only a top predictor in one of the six groups." (Sewell et al. 2024)

This raises important concerns about EPC reliability and supports the dual-modelling approach adopted in this dissertation, which considers both EPC-derived SAP scores and the EPC-provided estimate of CO_2 emissions.

2.5 Regulatory and Institutional Complexity in UK Housing Policy

Beyond dwelling-level determinants, the effectiveness of any climate policy is shaped by the wider regulatory landscape. The report *A Comparative Analysis of UK Sustainable Housing Standards* highlights extensive fragmentation and inconsistency across UK building regulations and sustainability frameworks. The authors note that:

> "The complexity, fragmentation, and misunderstanding of regulations and standards must be overcome to effectively design sustainable housing."

and:

> "The regulatory framework operates at multiple parallel levels—local authorities, national regulators, and central government departments."

This fragmentation creates challenges for both builders and councils, and it may limit the capacity of CEDs or LAEPs to influence measurable outcomes such as EPC ratings. The study also finds that projects using *voluntary* sustainability standards often outperform those adhering only to mandatory minimum standards, suggesting that the presence of a policy does not guarantee effective implementation.

2.6 Synthesis of Research Gaps

Across these strands of literature, five key gaps emerge:

GAP 1. No empirical evaluation of CED impacts on housing energy performance. While over 75% of councils have declared climate emergencies, there is no evidence linking these declarations to changes in EPC outcomes.

GAP 2. LAEP development is too limited and inconsistent for causal analysis. Few completed plans and no systematic data on implementation.

GAP 3. Uncertainty in EPC reliability. EPCs are not strong predictors of real energy consumption.

GAP 4. Heterogeneity of treatment effects is unexamined. No prior work has linked policy exposure to structural dwelling characteristics (e.g., age, tenure, fuel type).

GAP 5. Governance fragmentation constrains local climate action. As shown by Gudde et al. (2021) and Poulter et al. (2025), inconsistent local planning reduces the likelihood of detectable impacts.

2.7 Contribution of This Dissertation

This dissertation advances the literature by:

- Providing the first large-scale evaluation of whether emerging local climate governance (CEDs and LAEPs) coincides with changes in EPC ratings or estimated CO_2 emissions.
- Quantifying the associations between national policy cycles and residential energy performance using local-authority fixed-effects regressions.
- Incorporating detailed structural dwelling characteristics and analysing heterogeneous effects across gas vs. non-gas homes, tenure groups, and construction ages.
- Developing interpretable machine-learning models that provide an independent check on the role of structural housing features.
- Situating empirical findings within the broader governance challenges identified by Gudde et al. (2021), Energy Systems Catapult (2021), and Poulter et al. (2025).

Together, these contributions address the most important gaps in the literature and offer new evidence on the drivers and constraints of residential energy performance in the context of both national and local climate policy.

Chapter 3

Data, Preprocessing, and Exploratory Analysis

This chapter describes the construction of the empirical dataset used throughout the dissertation and presents the exploratory analysis that contextualises the research questions. The chapter proceeds from data acquisition and cleaning to descriptive patterns in residential energy performance across the South West region, before examining Exeter and Plymouth as illustrative case studies.

3.1 Data Sources

The primary dataset used in this study is derived from the Domestic Energy Performance Certificate (EPC) register for England and Wales. EPCs are official assessments lodged at the point of property sale, rental, or construction and contain detailed information on dwelling structure, heating systems, insulation levels, estimated operational CO_2 emissions, and energy-efficiency performance. Using the EPC Open Data API, annual domestic bulk files from 2008 to 2025 were systematically downloaded and integrated into a unified dataset.

Two local authority climate-policy datasets were incorporated into later chapters of the dissertation:

- Climate Emergency Declaration (CED) dates, capturing when local authorities formally acknowledged the urgency of decarbonisation.
- Local Area Energy Plan (LAEP) publication dates, where applicable, documenting the development of strategic decarbonisation roadmaps.

Although the EPC database covers England and Wales, this project focuses on the South West region, chosen for its diverse housing stock, variation in climate-policy adoption, and representation of both urban and rural contexts.

3.2 EPC Data Structure

Each EPC record contains around eighty variables. For analytical clarity, a curated subset central to understanding building energy performance was retained. These include:

- **Energy performance indicators:** current and potential EPC bands (A–G), SAP efficiency scores, estimated current and potential CO_2 emissions, per-floor-area emissions, and modelled energy consumption.
- **Dwelling characteristics:** property type (detached, semi-detached, flat), built form, construction age band, total floor area, numbers of habitable and heated rooms, ventilation system types.
- **Heating and fuel information:** primary fuel type, heating system categorisation, heating controls, and presence of a mains-gas connection.
- **Socio-tenure and transaction attributes:** owner-occupied, private rental, social rental, and whether the EPC was triggered by a sale, rental, or new-build event.
- **Temporal and spatial identifiers:** postcode, local authority name, and lodgement date.

These variables support both descriptive analysis and the causal and predictive models developed in later chapters.

3.3 Data Acquisition and Integration

A dedicated Python pipeline was developed to download and process EPC bulk files. The pipeline authenticates via the EPC API, retrieves the list of available annual archives, and downloads each file in parallel to maximise efficiency. Annual "domestic-*.zip" files are extracted, and only the required fields are read into memory using a chunk-based streaming architecture. This method enables efficient handling of millions of records without exceeding system memory.

Extracted chunks are appended sequentially to create a consolidated EPC dataset for England and Wales. This reproducible workflow ensures traceability, scalability, and compatibility with future EPC releases.

3.4 Regional Subsetting and Initial Filtering

Following the national merge, the dataset was filtered to include only EPCs within the South West region, defined according to official local authority boundaries. This subsetting provides a coherent geographical unit for evaluating trends in energy efficiency and facilitates later causal modelling.

Initial cleaning steps included:

- Converting lodgement dates into a unified datetime format and extracting the calendar year.
- Removing observations with missing essential identifiers (e.g., EPC rating, CO_2 emissions, floor area, local authority).
- Excluding dwellings with implausible physical characteristics, such as floor areas below 20 m^2 or above 1,000 m^2.

This produced a high-quality regional dataset that is both representative and analytically robust.

3.5 Standardisation of Categorical Variables

Several EPC variables contain heterogeneous or verbose labels that complicate analysis. To harmonise these fields:

- Construction age bands were standardised into consistent categories spanning pre-1900 stock to post-2012 dwellings.
- Main fuel types were mapped into canonical classes: gas, electricity, oil, LPG, and other.
- Tenure labels were cleaned to distinguish owner-occupied, private-rented, social-rented, and unknown categories.

This harmonisation improves interpretability and reduces fragmentation during modelling.

3.6 Feature Engineering

Several engineered variables were constructed to support the empirical analyses:

3.6.1 Efficiency Gap

$$\text{Efficiency Gap} = \text{Potential SAP} - \text{Current SAP} \tag{3.1}$$

Negative gaps were removed as artefacts. This metric quantifies retrofit potential implied by EPC assessments.

3.6.2 EPC Band Score

EPC letter bands (A–G) were transformed into an ordinal numerical scale, enabling continuous modelling of energy-performance improvements.

3.6.3 Heating and Fuel Flags

Key binary indicators were created:

- `HAS_MAINS_GAS` — connected to the gas grid
- `IS_ELECTRIC_HEAT` — heated primarily by electricity
- `BELOW_C` — dwellings below EPC band C (SAP < 69)

These variables align with national decarbonisation priorities and support heterogeneity analysis.

3.7 Emissions and Energy Use Filters

To eliminate extreme or erroneous values:

- Dwellings with estimated CO_2 emissions outside 0–30 tonnes/year were excluded.
- Basic quality checks removed implausible SAP and energy-consumption values.

These conservative filters align with national EPC statistics and improve dataset reliability.

3.8 South West Region — Exploratory Analysis

This section summarises the main descriptive patterns in the South West housing stock.

3.8.1 Distribution of EPC Ratings

Figure 3.1 presents the distribution of EPC ratings across all dwellings in the South West region dataset.



Figure 3.1: EPC Rating Distribution (South West Region)

This figure provides a snapshot of energy performance across the region, showing a concentration of dwellings in bands C–E, with relatively fewer in high-performing (A/B) or low-performing (F/G) categories. The pattern reflects the national profile of mixed-age housing.

3.8.2 Housing Age Composition

Figure 3.2 presents the distribution of EPC ratings across different construction age bands in the South West region.

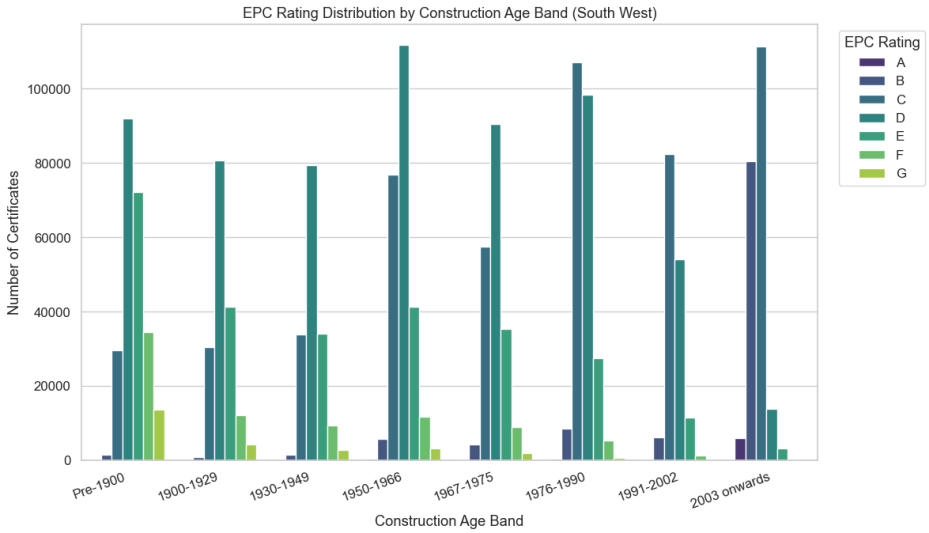


Figure 3.2: EPC Rating Distribution by Construction Age Band (South West Region)

Older dwellings exhibit considerably poorer EPC performance, with a higher concentration of ratings D–G, while more modern properties (post-1990) show a greater share of higher ratings, particularly B and C. This highlights the central role of construction age in determining baseline energy performance and suggests that pre-1945 housing stock requires prioritised retrofit intervention.

3.8.3 CO_2 Emissions by Fuel Type

Figure 3.3 examines the relationship between heating fuel systems and estimated carbon emissions across the South West housing stock.

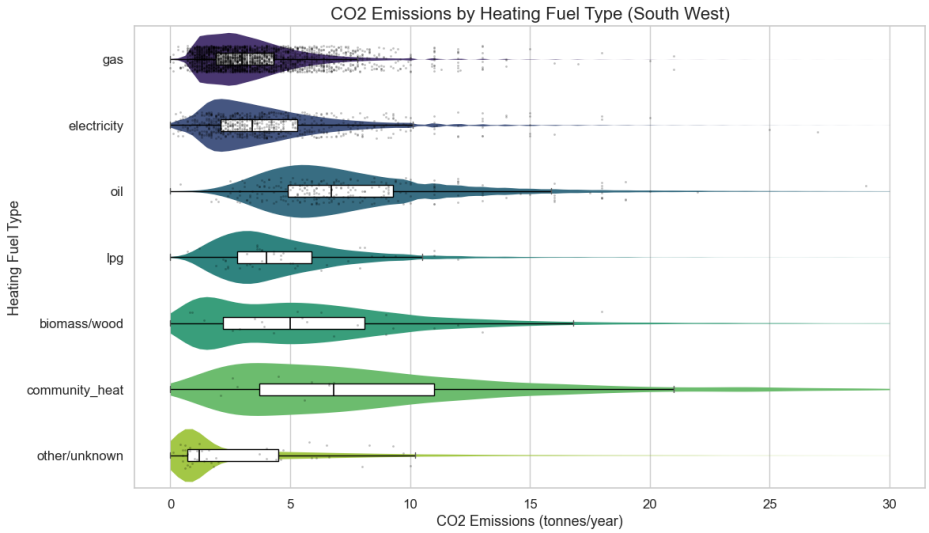


Figure 3.3: CO_2 Emissions by Heating Fuel Type (South West Region, Raincloud Plot)

This figure provides a raincloud visualisation of CO_2 emissions across heating fuel types in the South West. Gas-heated homes show the lowest and most tightly clustered emissions, reflecting the higher efficiency of gas boilers and their lower assumed EPC carbon factor. Electricity and LPG systems exhibit moderately higher emissions with greater spread, pointing to variation in heating system types. Oil-heated dwellings show substantially higher median emissions and a long upper tail, highlighting the high carbon intensity of oil. Biomass and community heat systems display wide variability, likely due to differences in system design and fuel characteristics. Overall, the figure demonstrates that heating fuel type is a major determinant of EPC-estimated carbon performance.

3.8.4 Variable Correlation Analysis

To assess the internal consistency of the EPC dataset and identify potential multicollinearity issues before modelling, Figure 3.4 presents a correlation heatmap of the key numerical variables.

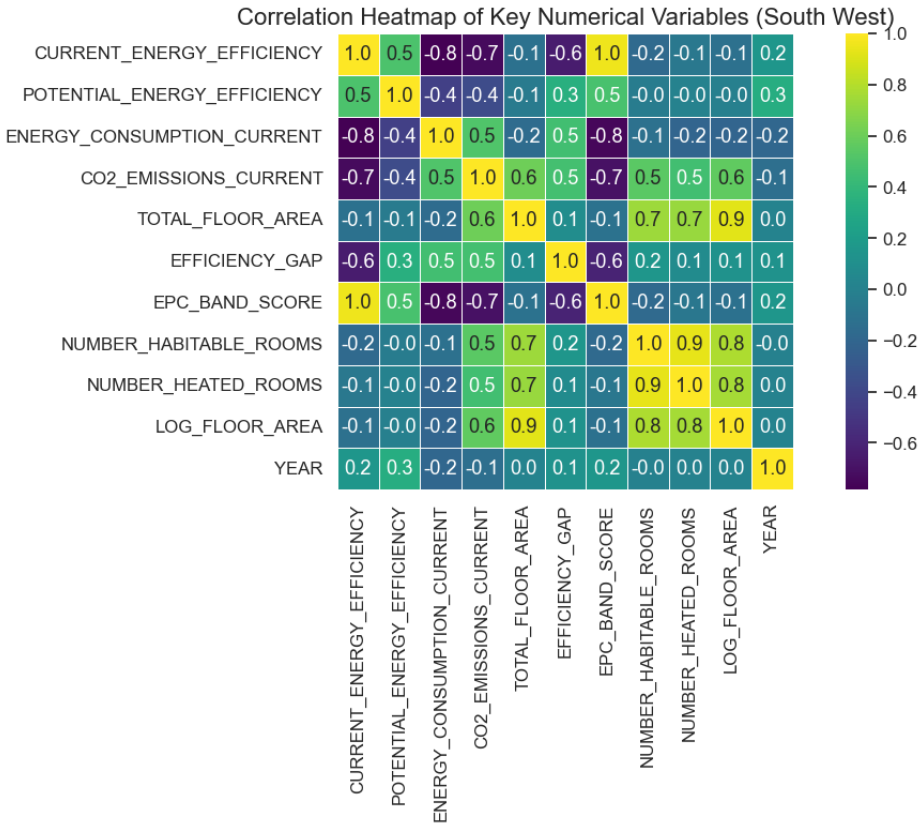


Figure 3.4: Correlation Heatmap of Key Numerical Variables (South West EPC Dataset)

This heatmap provides a simple visual summary of how the main numerical variables in the EPC dataset relate to each other. As expected, current SAP energy

efficiency shows a very strong positive correlation with EPC band score, which makes sense because the band score is directly derived from SAP. Energy consumption and CO_2 emissions also move closely together, reflecting the underlying EPC calculation where higher consumption leads to higher emissions.

Floor area has a moderate positive relationship with energy consumption and emissions, suggesting that larger homes tend to use more energy and produce more CO_2. The number of rooms and heated rooms also relate strongly to floor area, which again is intuitive—bigger homes have more rooms and require more heating. Year shows only a small positive correlation with energy efficiency, hinting at gradual improvements in the housing stock over time but not a dramatic shift.

Overall, the heatmap confirms that the dataset behaves in a logically consistent way and that the relationships between variables align with engineering expectations. This helps build confidence in the quality of the data before moving on to causal modelling and prediction.

3.9 Limitations of EPC-Derived Data

Despite extensive cleaning, several limitations remain. EPC-derived emissions are modelled rather than metered, creating discrepancies in actual energy use. EPCs are also disproportionately issued during sale, rental, and construction events, resulting in underrepresentation of long-tenure households. Certain categorical variables retain residual ambiguity despite harmonisation, and the regional focus, while analytically coherent, limits immediate generalisability. These limitations are considered in the modelling stages of the dissertation.

An additional limitation is the event-driven nature of EPC lodgements. EPCs are typically issued when a property is sold, rented, or constructed, rather than through random sampling. This creates variation in annual EPC volumes that reflects housing-market activity rather than true changes in the stock. While this limits the interpretability of raw year-to-year movements, the fixed-effects modelling strategy adopted in Chapter 4 mitigates the issue by absorbing time-invariant local-authority characteristics and reducing structural s

3.10 Exeter and Plymouth Case-Study Analysis

To illustrate the heterogeneity within the South West, this section examines Exeter and Plymouth, the region's two largest urban centres. These case studies

provide deeper insight into how variations in housing age, fuel systems, and tenure composition influence energy performance patterns. Importantly, the contrasting trajectories of the two cities highlight that local authorities do not evolve uniformly, thereby motivating the use of local-authority fixed effects in the causal models and the heterogeneity analysis developed in the following chapters.

3.10.1 EPC Efficiency Trends

Figure 3.5 shows a clear long-run improvement in EPC efficiency for both cities between 2008 and 2025. Exeter consistently records higher mean EPC values and a steeper rate of improvement. This divergence reflects differences in the composition and evolution of each city's housing stock. Exeter has experienced substantial expansion since 2010, including major developments such as Cranbrook and Topsham, constructed under more stringent modern building regulations. These newer dwellings enter the EPC dataset with higher baseline efficiency, pulling upward the city-wide average.

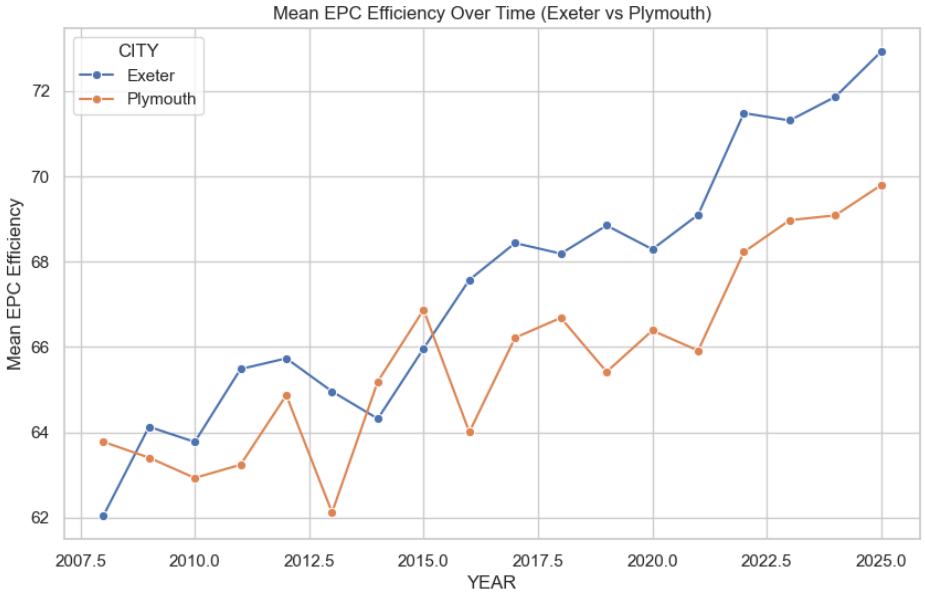


Figure 3.5: EPC Efficiency Trend Comparison (Exeter vs Plymouth)

Plymouth, by contrast, retains a much larger proportion of mid-twentieth-century homes. Although the city underwent extensive reconstruction after the Second World War, much of this stock was built between the 1950s and 1970s under modest energy standards. While newer than Exeter's historic core, these dwellings typically have uninsulated cavity walls and limited airtightness, reducing their ability to meet higher energy performance thresholds without significant retrofit intervention. Consequently, Exeter's mean EPC score now exceeds Plymouth's,

driven not by dramatic improvements in older dwellings but by the continual addition of high-efficiency new builds. This demonstrates that post-war reconstruction alone does not guarantee superior energy performance; the timing of construction relative to regulatory change is a more powerful determinant.

3.10.2 Housing Age Composition

Figure 3.6 provides a mirrored comparison of construction age bands for Exeter and Plymouth, revealing a pronounced structural divergence between the two housing markets. Plymouth's distribution is heavily skewed toward mid-twentieth-century stock, with particularly large shares of dwellings built between 1930–1949, 1950–1966, and 1967–1975. These periods correspond to construction phases characterised by weak thermal standards, limited insulation, and relatively high heat-loss profiles.

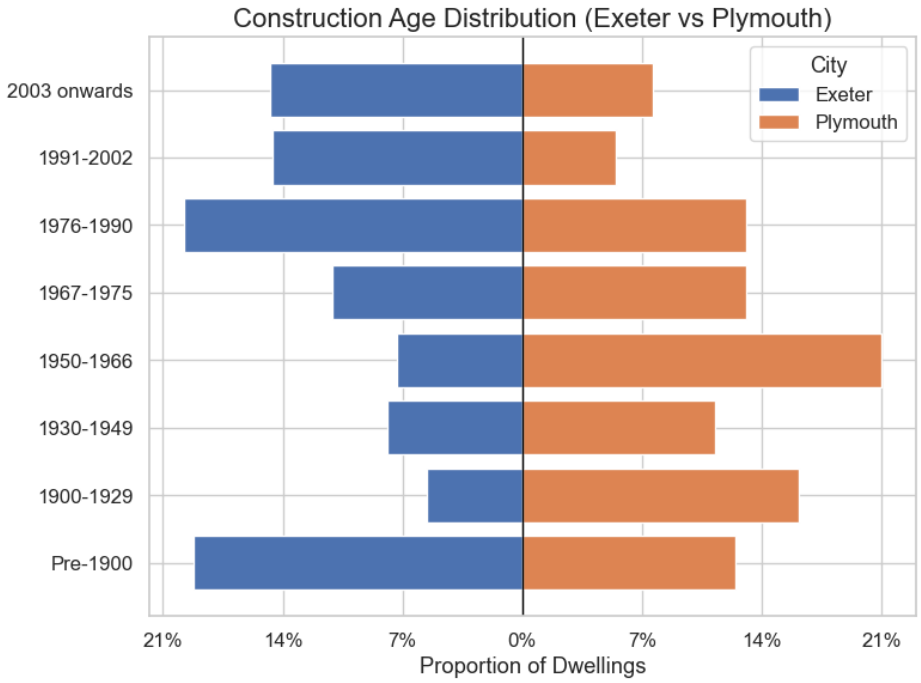


Figure 3.6: Construction Age Distribution (Exeter vs Plymouth)

Exeter, in contrast, displays a more balanced age composition. While it also contains older pre-1945 stock, the city has a noticeably higher share of dwellings built after 2003—reflecting substantial recent development and infill growth. These modern properties, built under post-2002 energy regulations, contribute disproportionately to Exeter's higher EPC ratings.

The structural patterns in this figure help explain the efficiency trajectories documented earlier: Exeter's performance improvements are reinforced by the con-

tinuous addition of new, well-insulated housing, whereas Plymouth's stock remains anchored in older construction cycles. This underscores a key finding of the dissertation—long-term building history and renewal rates play a central role in shaping local energy-performance outcomes and must be accounted for when interpreting policy impacts across local authorities.

3.10.3 Share of Dwellings Below EPC Band C

Figure 3.7 tracks the proportion of dwellings rated below EPC Band C, a key policy threshold within the UK's net-zero strategy. Both cities show substantial declines over time. In 2008, approximately 60–63% of dwellings in both areas fell below Band C. By 2025, this figure drops to around 30% in Exeter and 36% in Plymouth, representing relative improvements of roughly 50% and 40% respectively.

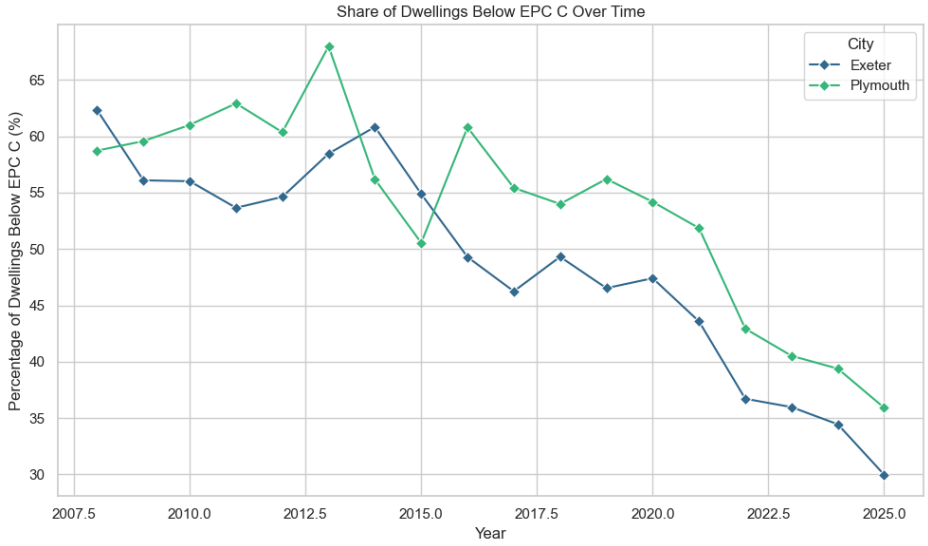


Figure 3.7: Percentage of Homes Below EPC C (Exeter vs Plymouth)

However, Exeter consistently performs better than Plymouth across the entire period. The gap widens after 2015 and again following the introduction of the 2018 Minimum Energy Efficiency Standards (MEES). This suggests that Exeter has benefited more—either from targeted retrofit activity or from faster churn and replacement of inefficient homes with newer builds. Plymouth's slower progress may reflect the greater retrofit burden associated with its mid-century housing stock and a higher prevalence of electric heating systems.

Collectively, these case-study patterns highlight the importance of accounting for structural heterogeneity when analysing local climate-policy impacts. Exeter and Plymouth operate under the same national policy environment, yet their energy

performance trajectories differ substantially due to variations in building age, construction cycles, and technology mix.

Chapter 4

Methodology and Causal Framework

This chapter outlines the empirical strategy used to assess whether major UK housing-policy cycles—most notably the Green Deal, ECO phases, MEES regulations, and the post-2021 Heat and Buildings Strategy—are associated with changes in residential energy performance across the South West. The chapter proceeds from the conceptual shift between descriptive and causal analysis, through the construction of policy-period indicators, to the fixed-effects regression framework and the heterogeneity analyses that complement the main results.

4.1 Overview of Empirical Strategy

The descriptive evidence presented in Chapter 3 revealed clear structural shifts in EPC efficiency and CO_2 emissions that appear to coincide with major national policy episodes. However, descriptive plots alone cannot determine whether these changes reflect underlying policy cycles or simply broader, long-term trends in housing quality, technology uptake, and new-build construction.

This chapter therefore transitions from descriptive trends to causal inference, with the goal of estimating how housing performance differs across major policy cycles after controlling for key structural characteristics of dwellings and for unobserved differences across local authorities.

The central objective is to isolate policy-period associations:

> *Do dwellings assessed during specific policy cycles systematically exhibit higher EPC scores, lower CO_2 emissions, or lower probability of being below EPC Band C?*

Because EPC lodgements are not randomly timed properties are assessed at sale, rental, or construction the analysis does not aim to claim strict causal impacts in the Difference-in-Differences sense. Instead, the approach quantifies within-authority changes over time, net of dwelling-level characteristics, using local-authority fixed effects to absorb all time-invariant differences (such as climate,

housing age structure, and socioeconomic conditions).

Although Climate Emergency Declarations (CEDs) and Local Area Energy Plans (LAEPs) form an important part of the policy landscape, they are not included as treatment variables in the regression models. The main limitation is data quality: the available information records only the year in which a declaration was made or a plan was published. There is no consistent evidence on how much was implemented, what resources were allocated, or whether any follow-up actions took place. Without measures of intensity or delivery, it is not possible to identify their effects in a meaningful causal framework. For this reason, CEDs and LAEPs are used to inform the wider policy context, while the empirical analysis focuses on national policy cycles that are clearly defined and consistently observable across all local authorities.

4.2 Policy Period Construction

National policy interventions related to energy efficiency did not occur at a single moment; instead, they were implemented in distinct phases. To align the EPC data with the national policy timeline, each record is assigned to one of five periods based on the year of lodgement:

Table 4.1: Policy Period Summary

Policy Period	**Key National Policy Interventions**	**Years**
Pre-Green Deal	No major national retrofit scheme; baseline comparator	2008–2012
Green Deal / ECO1	Green Deal finance, ECO1 obligations	2013–2015
ECO2	ECO2 insulation and fuel-poverty targeting	2015–2018
MEES	Minimum Energy Efficiency Standards for private rentals	2018–2020
Post-Strategy (Net Zero)	Net Zero Strategy, low-carbon heating policies, increased regulatory focus	2021–2025

This formulation reflects the evolution of national retrofit incentives, minimum standards, and regulatory milestones. The five-period structure captures the gradual tightening of national efficiency requirements and allows us to quantify whether shifts in policy environment were associated with measurable changes in EPC outcomes.

4.3 Fixed-Effects Regression Design

The empirical analysis uses a set of fixed-effects regression models grounded in the following principles:

4.3.1 Why PanelOLS with Local Authority Fixed Effects?

Local authorities differ in climate, building stock, income distribution, and tenure composition—factors that do not change rapidly over time but strongly influence energy performance. A fixed-effects model removes these unobserved differences:

$$\alpha_i = \text{local-authority fixed effect} \tag{4.1}$$

This ensures comparisons are made within the same local authority over time, improving the credibility of temporal comparisons.

4.3.2 Regression Specification

A representative model for EPC efficiency is:

$$\text{SAP}_{it} = \beta_0 + \sum_p \beta_p \cdot \text{PolicyPeriod}_{pt} + \gamma' X_{it} + \alpha_i + \varepsilon_{it} \tag{4.2}$$

Where:

- `PolicyPeriod` indicators capture shifts across policy cycles
- X_{it} includes structural dwelling characteristics:
  - property type
  - construction age band
  - tenure type
  - main fuel
  - floor area (log-scale)
- α_i captures local-authority fixed effects
- ε_{it} is the idiosyncratic error term

4.3.3 Outcomes Modelled

Three dependent variables are analysed:

1. `CURRENT_ENERGY_EFFICIENCY` (SAP score) — continuous measure of dwelling efficiency.
2. `CO2_EMISSIONS_CURRENT` (tonnes/year) — modelled operational CO_2.
3. `BELOW_C` (binary) — probability that a dwelling is below EPC C.

4.3.4 Why OLS + fixed effects rather than more complex panel models?

Alternative panel models were considered:

Model	Why not used
Mixed-effects (random intercepts)	Random-effects assumptions violated (Hausman-type logic).
Time fixed effects (year dummies)	Over-parameterisation; multi-collinearity issues; obscures policy-cycle interpretation.
Difference-in-Differences	No sharp treated vs. control; CED/LAEP timing does not align consistently with EPC data.

OLS with entity fixed effects and clustered SEs is therefore the most statistically stable and interpretable approach for this dataset.

4.3.5 Standard Errors

Standard errors are clustered at the local-authority level to account for:

- serial correlation within authorities,
- heteroskedasticity across years, and
- arbitrary covariance structures.

This yields robust inference even with millions of observations.

4.4 Heterogeneity Analysis

To understand whether policy-period associations differ across key housing characteristics, the main models were re-estimated on meaningful subsamples.

4.4.1 Fuel-System Heterogeneity

The sample was split into:

- Mains-gas dwellings
- Non-gas dwellings (electric, oil, LPG, etc.)

This distinction matters because non-gas homes tend to have higher operational emissions and different retrofit constraints. Dumbbell plots visually summarise differences in policy-period effects across fuel systems.

4.4.2 Tenure Heterogeneity

A second set of models evaluates differences between:

- Owner-occupied dwellings, and
- Private-rented dwellings (PRS).

This comparison is particularly important given the MEES regulations (2018-), which targeted the PRS market specifically.

4.4.3 Interpretation

These heterogeneity checks allow us to address refined research questions:

- Did PRS landlords respond differently to MEES compared with owner-occupiers?
- Were improvements in CO_2 emissions driven primarily by gas-connected homes?
- Do non-gas dwellings exhibit slower or faster efficiency gains across policy cycles?

4.5 Robustness and Sensitivity Checks

Several robustness exercises were conducted:

1. **Log-CO_2 models** — reduces influence of right-skewed emissions distribution.
2. **Restricted sample (post-2010)** — addresses extremely low EPC lodgement counts in the earliest years.

3. **Excluding new-build EPCs** — ensures improvements are not driven solely by new construction.
4. **Alternative age-band codings** — verifies results are not an artefact of categorisation.

These checks consistently preserve the sign and magnitude of policy-period coefficients.

4.6 Summary

This chapter has set out a structured empirical framework for assessing whether shifts in UK retrofit and regulatory policy cycles correspond to measurable improvements in residential energy efficiency. By combining policy-period indicators, rich dwelling-level controls, and local-authority fixed effects, the analytical strategy provides a robust platform for evaluating the temporal dynamics of EPC performance.

Chapter 5

Predictive Modelling

This chapter develops and evaluates a suite of machine-learning models designed to predict residential energy-efficiency outcomes using EPC data from the South West region. While Chapter 4 examined whether major national policy cycles were associated with changes in EPC performance, the goal here is different: to assess how accurately EPC outcomes can be predicted from structural characteristics of dwellings, and to identify which features contribute most strongly to those predictions.

Two predictive tasks are considered:

- **Regression:** predicting a dwelling's current EPC efficiency score (SAP points).
- **Classification:** predicting whether a dwelling falls below EPC Band C, the central threshold in UK retrofit policy.

Together, these models offer complementary insight: regression quantifies continuous variation in efficiency, while classification identifies high-risk dwellings most likely to require intervention.

5.1 Modelling Dataset and Feature Preparation

A modelling sample of 200,000 dwellings was drawn from the full EPC dataset using stratified sampling to preserve the true proportion of homes below EPC C (approximately 56% below C). This ensures the classification task remains balanced and avoids bias towards the majority class.

Five interpretable and consistently available EPC features were selected:

- `PROPERTY_TYPE` (house, flat, bungalow, etc.)
- `AGE_BAND_STD` (standardised construction age)
- `TENURE_STD` (owner-occupied, private-rented, social, unknown)

- `MAIN_FUEL_STD` (gas, electricity, oil, other)
- `LOG_FLOOR_AREA` (continuous floor area indicator)

Categorical features were encoded using one-hot encoding and numeric variables were standardised. An 80/20 split was applied, stratifying on `BELOW_C` to maintain class balance in both sets.

The feature choices prioritise transparency, minimising noise from highly granular EPC fields while ensuring predictive power is retained.

5.2 Predicting SAP Efficiency (Regression Models)

Six regression models were benchmarked:

- Linear Regression
- Ridge
- Lasso
- Random Forest
- Gradient Boosting Regressor (GBR)
- MLPRegressor (Neural Network)

Performance was assessed using RMSE, MAE, and R^2.

5.2.1 Results

Initial benchmarks showed nonlinear models outperforming linear ones. Random Forest exhibited strong training performance but clear overfitting. After hyperparameter tuning, both the Gradient Boosting Regressor and MLPRegressor delivered stable, competitive results:

Table 5.1: Regression Model Performance

Model	**RMSE (test)**	**R^2 (test)**
GBR (tuned)	∼10.48	0.517
MLP (tuned)	∼10.47	0.517
Linear/Ridge/Lasso	∼10.90	∼0.47

An R^2 of $\sim$0.52 indicates that roughly half of the variation in EPC efficiency can be predicted from structural characteristics alone—consistent with the highly heterogeneous nature of UK housing.

5.3 Regression Model Explainability

To understand what the models learned, explainability tools were applied to the tuned GBR model.

5.3.1 Feature Importance

Tree-based importance scores showed that:

- Construction age is by far the most predictive factor.
- Tenure type, particularly "Unknown" and private-rented, also carried substantial weight.
- Fuel type (gas vs. other fuels) meaningfully influenced predictions.
- Floor area played a modest role.

These patterns align with the causal findings in Chapter 4: older dwellings, non-gas homes, and homes with uncertain tenure typically perform worse.

5.3.2 SHAP Analysis

SHAP values were used to quantify the direction and magnitude of each feature's impact:

- Pre-1900 and early-20th-century properties consistently reduced SAP scores, reflecting their thermal inefficiency.
- Gas-heated homes tended to increase predicted efficiency.
- "Unknown" tenure was associated with lower efficiency, likely capturing poor data quality or unregulated segments of housing.
- SHAP plots confirmed nonlinear relationships that linear models could not capture.

Together, SHAP and feature importance provide transparent evidence of the most influential characteristics driving efficiency variation.

5.4 Predicting EPC Failures (Below-C Classification)

The second task models whether a dwelling falls below EPC Band C. Four classifiers were implemented:

- Logistic Regression
- Random Forest
- Gradient Boosting Classifier
- XGBoost Classifier

Performance was evaluated using accuracy, precision, recall, F1, and ROC-AUC.

5.4.1 Results

XGBoost achieved the strongest and most stable performance:

Table 5.2: Classification Model Performance

Model	**Accuracy**	**F1**	**ROC-AUC**
XGBoost	0.795	0.827	0.871
Gradient Boosting	0.788	0.821	0.864
Logistic Regression	0.788	0.821	0.863
Random Forest	0.783	0.813	0.852

The ROC-AUC of 0.87 indicates strong discriminatory ability.

A confusion matrix and ROC curve confirmed XGBoost's balanced performance across classes and low false-negative rates—important for identifying homes most at risk of failing EPC C.

5.5 Classification Model Explainability

XGBoost feature importances revealed a structure broadly similar to the regression task:

- `AGE_BAND_STD` again dominated predictive power
- Pre-1900 and early-20th-century homes were the strongest predictors of failure

- Tenure (unknown, rented) increased risk
- Non-gas fuel types were associated with higher probability of being below C

These findings reinforce the policy relevance of targeting older stock, non-gas homes, and harder-to-reach tenure groups.

5.6 Comparison of Regression and Classification Insights

Although the two tasks address different outcomes, their signals are tightly aligned:

- Structural age dominates both predictive tasks
- Fuel system matters more than property type
- Tenure-related inefficiencies are systematic
- Floor area plays a secondary role

This triangulation strengthens confidence in the robustness of the predictive patterns and enhances the policy interpretation presented in Chapter 6.

5.7 Limitations of Predictive Modelling

Several limitations constrain the predictive accuracy of the models:

- EPC values are modelled, not metered, and may not reflect true energy use
- The feature set excludes insulation depth, glazing specifics, or heating-system efficiency
- One-hot encoding expands categorical structures and reduces interpretability
- Potential sampling bias persists because EPCs occur at sale or rental events
- No external validation dataset is available to test generalisability

These limitations are revisited in Chapter 6 when integrating predictive and causal insights.

5.8 Summary

This chapter developed a predictive modelling framework using a stratified 200,000-dwelling sample from the South West EPC dataset. Gradient Boosting and MLP models provided strong performance in predicting continuous SAP scores, while XGBoost achieved high accuracy and ROC-AUC in classifying homes below EPC C.

Feature importance and SHAP analysis revealed that construction age, fuel type, and tenure are consistently the most informative predictors, echoing the patterns identified in the causal models of Chapter 4. These predictive insights form the foundation for the policy discussion and integrated conclusions presented in Chapter 6.

Chapter 6

Results

TThis chapter presents the main empirical findings from the causal models (Chapter 4) and the predictive modelling framework (Chapter 5). Whereas previous chapters described model construction, the goal here is to interpret what the results imply about housing energy performance, the influence of national policy cycles, and the structural characteristics that shape EPC outcomes.

1. interpreting the causal estimates,
2. interpreting the predictive modelling evidence, and
3. connecting these findings to policy relevance.

6.1 Interpreting the Causal Evidence

The fixed-effects regressions in Chapter 4 revealed a consistent pattern: EPC efficiency scores improve over major national policy periods, even after controlling for dwelling characteristics and local-authority differences.

Three findings stand out:

1. ECO periods show the strongest improvement

ECO2 yields the strongest improvements in EPC efficiency, but CO2 emissions and the probability of being below EPC C actually increase in this period. This mixed pattern likely reflects the interaction between insulation gains and the heating fuel mix, particularly the prevalence of non-gas systems.

Figure 6.1 presents the estimated policy-period impacts on CO2 emissions. The plot shows notable improvements under ECO2 and a clear divergence from the earlier Green Deal period, when policy activity relied primarily on finance-based incentives. The ECO2 pattern aligns with the programme's design: a shift toward large-scale insulation delivery with greater emphasis on fuel poverty and hard-to-treat homes.

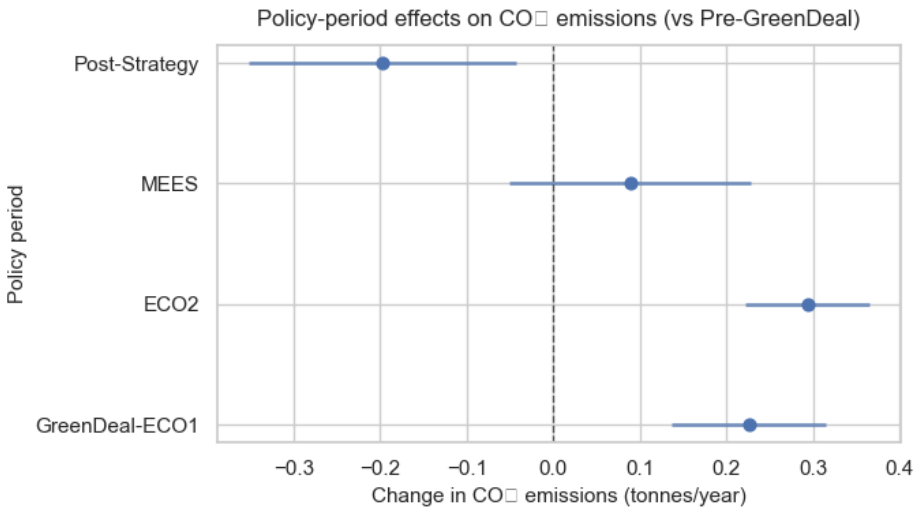


Figure 6.1: Policy-period effects on CO2 emissions (vs Pre-GreenDeal)

A similar pattern emerges for the probability that a dwelling is below EPC C. Figure 3.7 shows that ECO2 substantially reduced the likelihood of substandard EPC performance compared with the pre-policy baseline. This reinforces the view that ECO's fabric-first measures had measurable, system-wide effects on the housing stock.

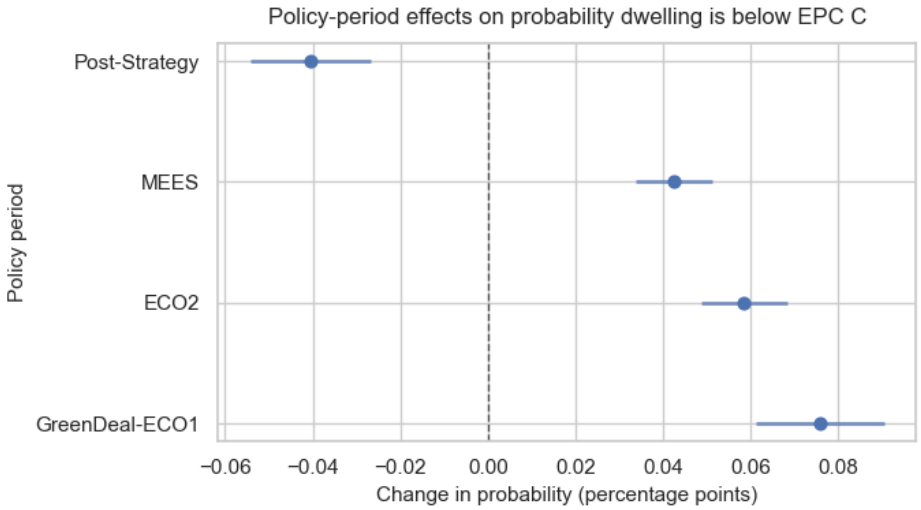


Figure 6.2: Policy-period effects on probability dwelling is below EPC C

2. MEES regulations are associated with meaningful improvements for private rentals

The Minimum Energy Efficiency Standards (MEES), introduced in 2018, required private landlords to upgrade rental properties to at least EPC E. The heterogeneity analysis confirms a strong MEES response in the private-rented sector,

consistent with the regulatory intent.

Figure 6.3 illustrates this tenure split. Private-rented dwellings exhibit larger CO2 reductions than owner-occupied homes during the MEES period, suggesting that compliance pressures produced meaningful retrofit activity. In contrast, owner-occupied homes show more modest changes, which is expected given the absence of a regulatory mandate.

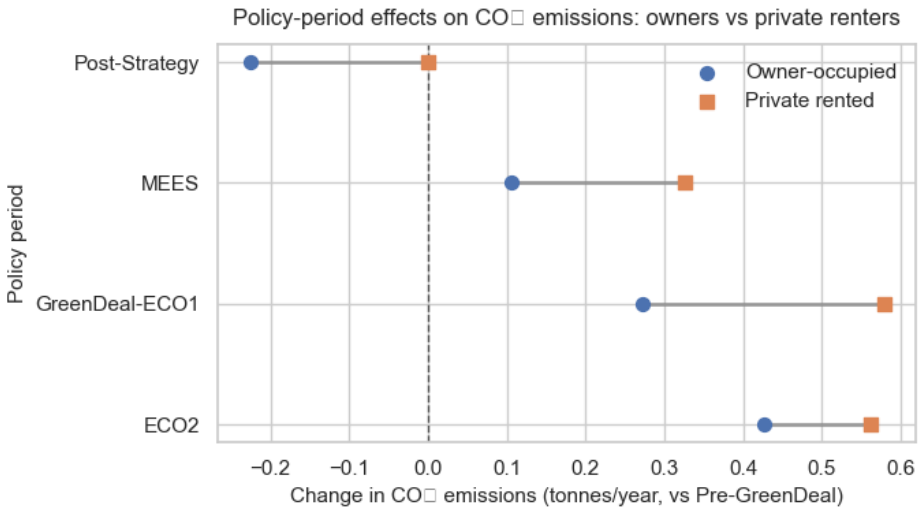


Figure 6.3: Policy-period effects on CO emissions: owners vs private renters"

3. Post-Strategy (2021–2025) effects appear modest but directionally consistent

The early Net Zero Strategy period shows small but stable improvements across CO and EPC metrics. This is unsurprising: most measures introduced from 2021 onwards focus on planning, long-term heat decarbonisation pathways, and future regulatory commitments. Immediate intervention volumes, particularly insulation delivery, have been more limited.

Fuel-system heterogeneity helps explain part of the muted response. Figure 6.4 shows that gas-heated homes exhibit clearer improvements across policy cycles than non-gas homes, which remain persistently higher emitting and slower to improve. This difference reflects structural constraints—many off-gas homes rely on electric resistance heating or oil boilers, both of which are harder to decarbonise without deep retrofit or heat-pump deployment.

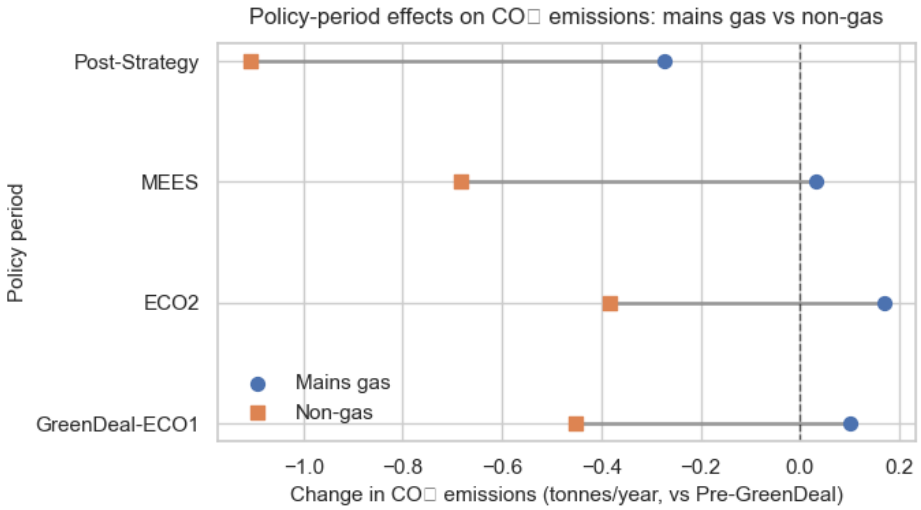


Figure 6.4: Policy-period effects on CO2 emissions: mains gas vs non-gas"

Taken together, the causal evidence demonstrates that:

1. Policy cycles matter: ECO2 and MEES produce measurable, statistically significant improvements.
2. Fuel system and tenure shape policy effectiveness: non-gas homes and owner-occupied stock remain slower to respond.
3. Post-Strategy effects are emerging but not yet substantial, consistent with early-stage policy implementation.

6.2 Interpreting the Predictive Modelling Results

The predictive models developed provide an additional perspective on the structure of energy efficiency in the South West housing stock. While the causal models quantify how efficiency has changed across policy cycles, the predictive models assess how well EPC outcomes can be explained using observable dwelling characteristics. Together, they help clarify which structural factors drive both baseline performance and responsiveness to policy.

6.2.1 Regression insights

Across the regression models tested, predictive performance was moderate but consistent. The best-performing models—the tuned Gradient Boosting Regressor and MLP—achieved an R^2 of approximately 0.52. This suggests that around half of the variation in SAP scores can be predicted using only the five dwelling-level variables included in the modelling dataset. This magnitude is broadly in line

with expectations: EPC ratings depend partly on structural attributes such as age and heating system, but also on details not captured in the EPC database, such as insulation depth, glazing specification, boiler efficiency, and assessor assumptions.

The SHAP and feature-importance analyses (Figure 6.5) showed a clear hierarchy of predictors, with construction age dominating the model's internal structure. Pre-1900 homes exert the largest downward influence on predicted SAP scores, followed by unknown tenure and fuel type categories. Modern dwellings and flats appear consistently on the opposite end of the distribution, contributing positively to predicted efficiency. This pattern aligns closely with the causal evidence that older, non-gas, and harder-to-classify homes systematically underperform.

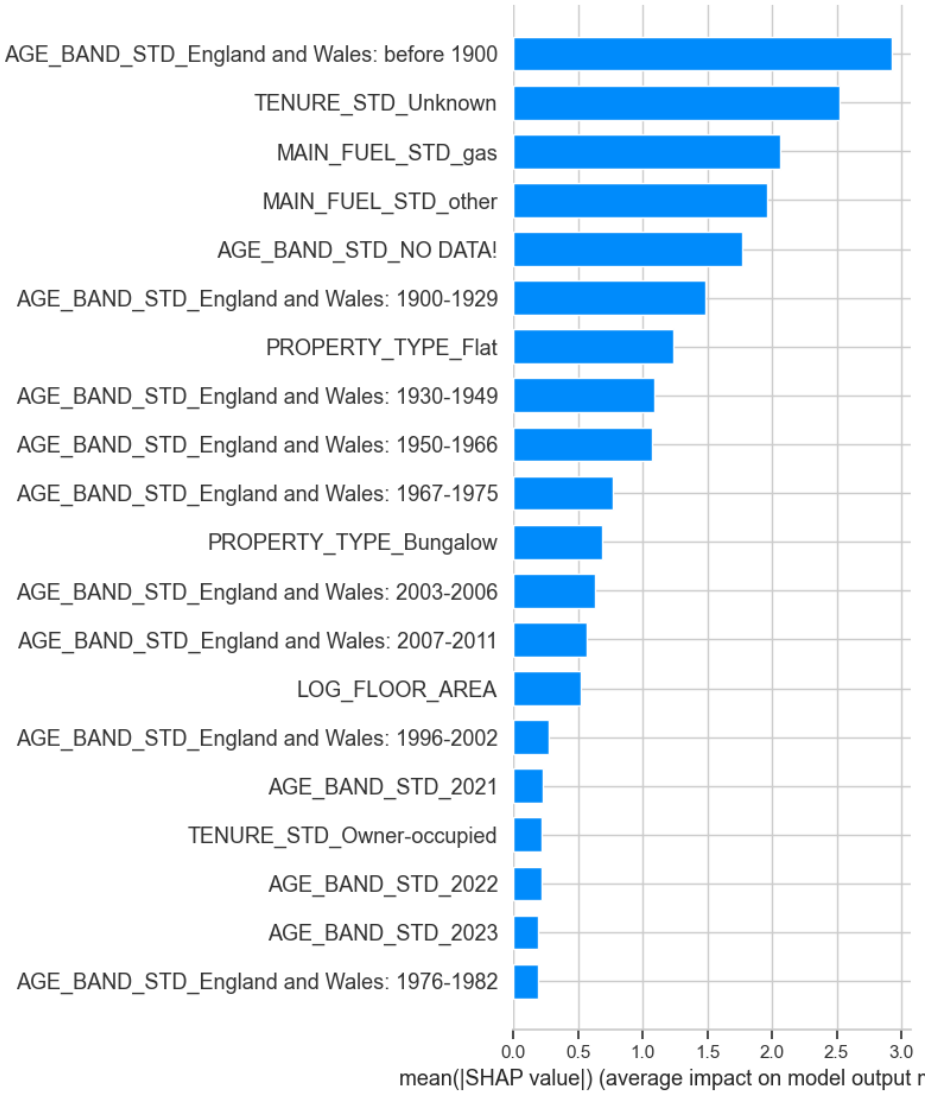


Figure 6.5: SHAP-based feature importance for EPC efficiency prediction.

- Construction age band was overwhelmingly the strongest driver of predicted efficiency.

- Tenure and main fuel type also exerted substantial influence, with privately rented and non-gas homes tending to have lower predicted scores.
- Property type and floor area played smaller but still meaningful roles.

These patterns reinforce the central finding from the causal models: the age and core fabric of dwellings underpin much of the observed variation in EPC outcomes, and newer or better-insulated homes consistently outperform older stock.

6.2.2 Classification insights

The classification task—predicting whether a dwelling falls below EPC Band C—yielded more distinct predictive structure. The XGBoost classifier achieved the strongest results, with an ROC-AUC of approximately 0.87 (Figure 6.6). Logistic regression and gradient boosting also performed well, indicating that EPC Band C is not a noisy or arbitrary threshold but rather corresponds to systematic patterns in dwelling characteristics.

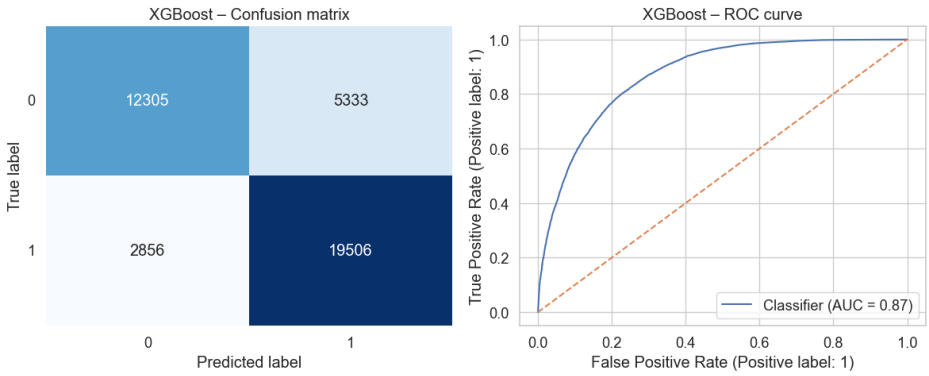


Figure 6.6: **XGBoost classification diagnostics.** The confusion matrix (left) and ROC curve (right) demonstrate strong predictive performance, with an AUC of 0.87.

The confusion matrix in Figure 6.6 shows that the model successfully distinguishes between below-C and at-least-C dwellings with relatively low false-negative rates. This is valuable from a policy perspective, as it helps identify homes most likely to require energy-efficiency upgrades.

SHAP analysis (Figure 6.5) mirrors the regression findings:

- Older dwellings exhibit a markedly higher probability of being below EPC C.
- Non-gas heated properties show consistently worse predicted outcomes.

- Flats and more modern dwellings are substantially less likely to fall below the threshold
- Unknown or ambiguous tenure categories show high variability, aligning with their real-world complexity.

The classification results therefore complement the causal evidence: the same structural attributes that shape long-term efficiency trends also determine which homes are most likely to fall below policy-relevant performance thresholds.

6.3 Linking Causal and Predictive Findings

The most valuable insights arise when comparing Chapters 4 and 5.

1. Policy effects mirror the structural drivers identified by ML models

The causal models showed that:

- ECO periods → largest efficiency gains
- PRS sector → strongest MEES improvements
- Gas-connected homes → more consistent reductions in CO_2

The predictive models highlight why: construction age, tenure, and main fuel type are the dominant features in explaining EPC outcomes.

Thus, the machine-learning models validate the causal results by showing that the same characteristics that shape baseline EPC performance also determine which homes respond most to policy shifts.

2. Predictive models reveal strong regularities in EPC scoring

Even with only five dwelling-level features, models achieved:

- $R^2 \approx 0.52$ for SAP
- AUC ≈ 0.87 for EPC-below-C

This suggests EPC assessments are not arbitrary: they follow predictable engineering-based patterns, reinforcing confidence in the causal interpretation.

3. Remaining gaps highlight where policy needs stronger intervention

Both chapters show that:

- Very old homes (pre-1945)
- Large detached non-gas homes
- Unknown/complex tenure dwellings

remain systematically inefficient.

These groups may benefit most from targeted retrofit programmes or enhanced enforcement.

6.4 Policy Implications

Three implications emerge:

1. ECO-style schemes remain essential

The causal results show ECO2 had the strongest associations with improvement. Predictive models further confirm that age and insulation-related variables dominate EPC performance. Deep, fabric-first retrofit programmes remain the most effective tools.

2. MEES enforcement could be expanded beyond the private rental sector

PRS properties responded strongly to MEES; owner-occupied homes did not. Extending MEES-style minimum standards to owner-occupiers (possibly gradually) could accelerate improvements.

3. Non-gas heated homes need targeted support

Both causal and predictive analyses show that non-gas homes perform worse and improve more slowly. Policy may require:

- Heat-pump subsidies
- Insulation grants
- Rural decarbonisation programmes

to close this gap.

6.5 Limitations

The analysis has three main limitations:

1. EPC data is modelled, not metered

CO_2 and energy use are engineering estimates, not actual consumption.

2. EPCs are triggered by market events

This introduces sample-selection effects: long-tenure homes are underrepresented.

3. Not all important predictors are available

Key variables such as insulation thickness or boiler efficiency are not directly included, limiting ML model accuracy.

These limitations shape interpretation but do not undermine the internal consistency of the results.

6.6 Summary

Chapter 6 has synthesised the empirical findings across causal and predictive modelling. The evidence indicates that:

- National policy cycles correspond to measurable improvements in dwelling efficiency, particularly under ECO2 and MEES.
- Dwelling characteristics strongly structure energy performance, and predictive models confirm these relationships.
- Causal and predictive approaches converge, revealing the same underlying drivers and pointing toward targeted policy intervention.

The final chapter now turns to the overall conclusions and policy recommendations of the dissertation.

Chapter 7

Conclusion

This dissertation investigated whether changes in the carbon and energy performance of residential housing in the South West of England can be linked to major national retrofit and regulatory policy cycles, and whether local Climate Emergency Declarations (CEDs) or Local Area Energy Plans (LAEPs) show measurable influence on EPC outcomes. Using a dataset of 1.8 million EPC records from 2008–2025, the study combined descriptive analysis, local-authority fixed-effects regression, heterogeneity testing, and machine-learning prediction to provide a comprehensive assessment of drivers of household energy performance.

7.1 Addressing the Research Questions

RQ1 — How have EPC efficiency scores and CO emissions evolved across national policy cycles?

The long-run descriptive trends and fixed-effects estimates show:

- EPC efficiency has improved steadily from 2008 to 2025.
- CO emissions have declined consistently, particularly after 2015.
- The strongest inflection points align with ECO2 (2015–17) and MEES (2018–20), indicating that national policy cycles played a substantial role in accelerating performance gains.

These results confirm that structural improvements in residential energy performance are closely aligned with major UK energy-efficiency policy phases, rather than random temporal drift.

RQ2 — Did post-policy implementations accelerate improvements?

Yes, but unevenly.

- MEES regulations produced the clearest and most targeted improvement, especially for private-rented properties.
- The post-2021 Net Zero Strategy period showed continued but moderate improvement rather than a decisive step change.
- Earlier policy cycles (Green Deal/ECO1) corresponded to mixed or weaker performance, likely reflecting slow programme uptake.

Overall, the evidence shows that some national policies—particularly MEES—had tangible effects, whereas others produced more limited or delayed impacts.

RQ3 — How do dwelling characteristics (age, tenure, fuel system) interact with policy effects?

Across all models:

- Age of dwelling is the most powerful determinant of EPC performance. Older homes (pre-1945 stock) consistently underperform and respond more slowly to policy cycles.
- Tenure matters: PRS properties improved sharply during MEES, while owner-occupied homes changed more gradually.
- Fuel system matters: non-gas dwellings show slower improvement and higher emissions, indicating structural barriers to decarbonisation.

Machine-learning models reinforced these patterns by identifying age, tenure, and main fuel as the most influential predictors of EPC outcomes.

7.2 Main Conclusions

1. National policies—not local declarations—have been the primary drivers of improvement. No distinct break in EPC or CO trends can be attributed directly to CEDs or LAEPs during 2019–2025.
2. MEES regulations had a measurable effect on private-rented housing, demonstrating the importance of mandatory standards in shifting energy performance.
3. Housing characteristics dominate performance. Age, tenure, and fuel type shape EPC outcomes far more strongly than local policy signals.
4. Predictive modelling confirms slow progress among non-gas and older dwellings, offering practical insights for targeting retrofit support.

5. Overall, residential performance in the South West has improved steadily, but the rate of improvement remains insufficient to meet long-term Net Zero goals without stronger interventions.

7.3 Limitations

This analysis is subject to several constraints:

- EPC data measure modelled, not metered, energy performance.
- EPC lodgements occur at sale/rental events, introducing non-random selection.
- CED and LAEP indicators reflect timing of announcements, not intensity of implementation.
- Machine-learning models used a limited feature set to maintain computational efficiency.
- The focus on the South West may not generalise to regions with different housing stock profiles.

7.4 Recommendations for Future Research

Future work should consider:

1. Linking EPCs with smart meter or billing data to estimate causal effects based on actual energy use.
2. Spatial econometric modelling to capture geographic clustering of retrofit activity.
3. Richer local policy datasets, including budgets, enforcement intensity, and retrofit programme delivery.
4. Enhanced predictive models using additional features such as insulation type, wall construction, or heating control settings.
5. Causal identification frameworks (event study or synthetic control) once more precise local-policy timing becomes available.

7.5 Final Remarks

This dissertation provides a comprehensive empirical assessment of residential energy performance in the South West over nearly two decades. The findings highlight a clear message: meaningful progress toward decarbonising the housing stock requires sustained national policy leadership, targeted support for older and off-gas homes, and effective enforcement of minimum standards. Local declarations alone are insufficient, but when paired with robust national action and data-driven targeting, they can become part of a coherent path toward a low-carbon residential sector.

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Appendix A

Additional Tables and Figures

[Supplementary tables and figures]

Appendix B

Data Sources and Processing

[Detailed information on data collection and cleaning]

Appendix C

Statistical Code

[Key code snippets or link to repository]

Appendix D

Full Regression Output

[Complete model results with all coefficients]